

[Novel approaches; improved diagnostics and therapeutics with DNA microarrays. II. Applications].

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The most important (future) applications of DNA microarrays in medicine are based on three different molecular analyses: gene expression analysis, genotyping with a specific type of microarray (single nucleotide polymorphism chip (SNP chip)), and DNA sequencing to search for a well-known mutation in a certain gene. With gene expression analysis one can obtain a 'molecular signature' of a tissue by allowing tissue RNA to react with a DNA microarray. This information may help to refine disease classifications, to guide the choice of therapy and to find new therapeutic targets. Genotyping utilizes genomic DNA that, after digestion, reacts with a SNP chip to obtain an individual SNP pattern. An SNP is a change of a single base pair within a gene sequence that can sometimes influence the function of the gene product. SNPs are considered to be variations in the genome and should be distinguished from mutations. These variations can for instance provide information about the probability of a certain disease, or the effectiveness or side effect of a certain drug. Sequencing well characterized genes to search for mutations is used to screen for genetic abnormalities. This process can be accelerated with the aid of oligonucleotide arrays. With this tool an individual's risk of developing certain forms of cancer or monogenetic diseases can be estimated.

Identification of SNPs, or mutations in sequence chromatograms.

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With the completion of the human genome sequencing project in 2001, the identification of novel markers is rapidly gaining importance. It is increasingly recognized that SNPs (single nucleotide polymorphisms) are good markers for disease susceptibility. SNPs are DNA sequence variations that occur when a single nucleotide in the genome sequence is altered in at least 1 % of the population. SNPs may have no effect on cell function, but scientists believe that they could predispose people to disease or influence their response to a drug. This chapter describes the method of using fluorescent based sequencing to detect SNPs and mutations. Sequencing provides information on the type and location of the SNPs with high accuracy. Researchers will need to provide information on the area of the genome they wish to sequence to design primers to PCR amplify the specific region.

Identification and In-Silico study of non-synonymous functional SNPs in the human SCN9A gene.

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Single nucleotide polymorphisms are the most common form of DNA alterations at the level of a single nucleotide in the genomic sequence. Genome-wide association studies (GWAS) were carried to identify potential risk genes or genomic regions by screening for SNPs associated with disease. Recent studies have shown that SCN9A comprises the NaV1.7 subunit, Na⁺ channels have a gene encoding of 1988 amino acids arranged into 4 domains, all with 6 transmembrane regions, and are mainly found in dorsal root ganglion (DRG) neurons and sympathetic ganglion neurons. Multiple forms of acute hypersensitivity conditions, such as primary erythralgia, congenital analgesia, and paroxysmal pain syndrome have

been linked to polymorphisms in the SCN9A gene. Under this study, we utilized a variety of computational tools to explore out nsSNPs that are potentially damaging to health by modifying the structure or activity of the SCN9A protein. Over 14 potentially damaging and disease-causing nsSNPs (E1889D, L1802P, F1782V, D1778N, C1370Y, V1311M, Y1248H, F1237L, M936V, I929T, V877E, D743Y, C710W, D623H) were identified by a variety of algorithms, including SNPnexus, SNAP-2, PANTHER, PhD-SNP, SNP & GO, I-Mutant, and ConSurf. Homology modeling, structure validation, and protein-ligand interactions also were performed to confirm 5 notable substitutions (L1802P, F1782V, D1778N, V1311M, and M936V). Such nsSNPs may become the center of further studies into a variety of disorders brought by SCN9A dysfunction. Using in-silico strategies for assessing SCN9A genetic variations will aid in organizing large-scale investigations and developing targeted therapeutics for disorders linked to these variations.

Joint annotation of coding and non-coding single nucleotide polymorphisms and mutations in the SNPeffect and PupaSuite databases.

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Single nucleotide polymorphisms (SNPs) are, together with copy number variation, the primary source of variation in the human genome. SNPs are associated with altered response to drug treatment, susceptibility to disease and other phenotypic variation. Furthermore, during genetic screens for disease-associated mutations in groups of patients and control individuals, the distinction between disease causing mutation and polymorphism is often unclear. Annotation of the functional and structural implications of single nucleotide changes thus provides valuable information to interpret and guide experiments. The SNPeffect and PupaSuite databases are now synchronized to deliver annotations for both non-coding and coding SNP, as well as annotations for the SwissProt set of human disease mutations. In addition, SNPeffect now contains predictions of Tango2: an improved aggregation detector, and Waltz: a novel predictor of amyloid-forming sequences, as well as improved predictors for regions that are recognized by the Hsp70 family of chaperones. The new PupaSuite version incorporates predictions for SNPs in silencers and miRNAs including their targets, as well as additional methods for predicting SNPs in TFBSs and splice sites. Also predictions for mouse and rat genomes have been added. In addition, a PupaSuite web service has been developed to enable data access, programmatically. The combined database holds annotations for 4,965,073 regulatory as well as 133,505 coding human SNPs and 14,935 disease mutations, and phenotypic descriptions of 43,797 human proteins and is accessible via <http://snpeffect.vib.be> and <http://pupasuite.bioinfo.cipf.es/>.

Sensitive measurement of single-nucleotide polymorphism-induced changes of RNA conformation: application to disease studies.

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Single-nucleotide polymorphisms (SNPs) are often linked to critical phenotypes such as diseases or responses to vaccines, medications and environmental factors. However, the specific molecular mechanisms by which a causal SNP acts is usually not obvious. Changes in RNA secondary structure emerge as a possible explanation necessitating the development of methods to measure the impact of single-nucleotide variation on RNA structure. Despite the recognition of the importance of considering the

changes in Boltzmann ensemble of RNA conformers in this context, a formal method to perform directly such comparison was lacking. Here, we solved this problem and designed an efficient method to compute the relative entropy between the Boltzmann ensembles of the native and a mutant structure. On the basis of this theoretical progress, we developed a software tool, remuRNA, and investigated examples of its application. Comparing the impact of common SNPs naturally occurring in populations with the impact of random point mutations, we found that structural changes introduced by common SNPs are smaller than those introduced by random point mutations. This suggests a natural selection against mutations that significantly change RNA structure and demonstrates, surprisingly, that randomly inserted point mutations provide inadequate estimation of random mutations effects. Subsequently, we applied remuRNA to determine which of the disease-associated non-coding SNPs are potentially related to RNA structural changes.

AutoSNPdb: an annotated single nucleotide polymorphism database for crop plants.

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Single nucleotide polymorphisms (SNPs) may be considered the ultimate genetic marker as they represent the finest resolution of a DNA sequence (a single nucleotide), are generally abundant in populations and have a low mutation rate. Analysis of assembled EST sequence data provides a cost-effective means to identify large numbers of SNPs associated with functional genes. We have developed an integrated SNP discovery pipeline, which identifies SNPs from assembled EST sequences. The results are maintained in a custom relational database along with EST source and annotation information. The current database hosts data for the important crops rice, barley and Brassica. Users may rapidly identify polymorphic sequences of interest through BLAST sequence comparison, keyword searches of annotations derived from UniRef90 and GenBank comparisons, GO annotations or in genes corresponding to syntenic regions of reference genomes. In addition, SNPs between specific varieties may be identified for targeted mapping and association studies. SNPs are viewed using a user-friendly graphical interface. The database is freely accessible at <http://autosnpdb.qfab.org.au/>.

Clinical genotyping: the need for interrogation of single nucleotide polymorphisms and mutations in the clinical laboratory.

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Detection of single nucleotide polymorphisms (SNPs) and gene mutations is becoming more routine to the clinical laboratory. Completion of the Human Genome Project has led to new scientific knowledge of human disease processes that has revealed the most fundamental of abnormalities in nucleic acids while at the same time bringing some of the most sophisticated diagnostic tools to the clinical laboratory. In addition, public awareness (both lay persons and healthcare providers) and sensitivity to human genetics has increased tremendously. Together, this rapidly evolving science and increased public education has led to an increasing demand for genotypic testing. There are several clinical applications of human genotyping that are available using these newer technologies.

Revealing the human mutome.

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The number of known mutations in human nuclear genes, underlying or associated with human inherited disease, has now exceeded 100,000 in more than 3700 different genes (Human Gene Mutation Database). However, for a variety of reasons, this figure is likely to represent only a small proportion of the clinically relevant genetic variants that remain to be identified in the human genome (the 'mutome'). With the advent of next-generation sequencing, we are currently witnessing a revolution in medical genetics. In particular, whole-genome sequencing (WGS) has the potential to identify all disease-causing or disease-associated DNA variants in a given individual. Here, we use examples of recent advances in our understanding of mutational/pathogenic mechanisms to guide our thinking about possible locations outwith gene-coding sequences for those disease-causing or disease-associated variants that are likely so often to have been overlooked because of the inadequacy of current mutation screening protocols. Such considerations are important not only for improving mutation-screening strategies but also for enhancing the interpretation of findings derived from genome-wide association studies, whole-exome sequencing and WGS. An improved understanding of the human mutome will not only lead to the development of improved diagnostic testing procedures but should also improve our understanding of human genome biology.

Application of Nanotechnology for Sensitive Detection of Low-Abundance Single-Nucleotide Variations in Genomic DNA: A Review.

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Single-nucleotide polymorphisms (SNPs) are the simplest and most common type of DNA variations in the human genome. This class of attractive genetic markers, along with point mutations, have been associated with the risk of developing a wide range of diseases, including cancer, cardiovascular diseases, autoimmune diseases, and neurodegenerative diseases. Several existing methods to detect SNPs and mutations in body fluids have faced limitations. Therefore, there is a need to focus on developing noninvasive future polymerase chain reaction (PCR)-free tools to detect low-abundant SNPs in such specimens. The detection of small concentrations of SNPs in the presence of a large background of wild-type genes is the biggest hurdle. Hence, the screening and detection of SNPs need efficient and straightforward strategies. Suitable amplification methods are being explored to avoid high-throughput settings and laborious efforts. Therefore, currently, DNA sensing methods are being explored for the ultrasensitive detection of SNPs based on the concept of nanotechnology. Owing to their small size and improved surface area, nanomaterials hold the extensive capacity to be used as biosensors in the genotyping and highly sensitive recognition of single-base mismatch in the presence of incomparable wild-type DNA fragments. Different nanomaterials have been combined with imaging and sensing techniques and amplification methods to facilitate the less time-consuming and easy detection of SNPs in different diseases. This review aims to highlight some of the most recent findings on the aspects of nanotechnology-based SNP sensing methods used for the specific and ultrasensitive detection of low-concentration SNPs and rare mutations.

SNPs: impact on gene function and phenotype.

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Single nucleotide polymorphism (SNP) is the simplest form of DNA variation among individuals. These simple changes can be of transition or transversion type and they occur throughout the genome at a frequency of about one in 1,000 bp. They may be responsible for the diversity among individuals, genome evolution, the most common familial traits such as curly hair, interindividual differences in drug response, and complex and common diseases such as diabetes, obesity, hypertension, and psychiatric disorders. SNPs may change the encoded amino acids (nonsynonymous) or can be silent (synonymous) or simply occur in the noncoding regions. They may influence promoter activity (gene expression), messenger RNA (mRNA) conformation (stability), and subcellular localization of mRNAs and/or proteins and hence may produce disease. Therefore, identification of numerous variations in genes and analysis of their effects may lead to a better understanding of their impact on gene function and health of an individual. This improved knowledge may provide a starting point for the development of new, useful SNP markers for medical testing and a safer individualized medication to treat the most common devastating disorders. This will revolutionize the medical field in the future. To illustrate the effect of SNPs on gene function and phenotype, this minireview focuses on evidences revealing the impact of SNPs on the development and progression of three human eye disorders (Norrie disease, familial exudative vitreoretinopathy, and retinopathy of prematurity) that have overlapping clinical manifestations.
